25901090 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBCH202 - METABOLISM OF CARBOHYDRATES, PROTEINS AND LIPIDS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice questions

Answer all questions

Each question carries **1** marks

1. The only source of energy in erythrocytes is: [U] / [CO1]

   a). Beta Oxidation b). Krebs Cycle

   c). TCA Cycle d). Glycolysis

2. Select the product formed from the ketogenic amino acids: [U] / [CO3]

   a). Pyruvate b). Oxaloacetate

   c). Acetyl coA d). Alpha keto glutarate

3. What is the role of acyl CoA dehydrogenase enzyme in oxidation of palmitate? [U] / [CO4]

   a). oxidation b). Reduction

   c). transesterification d). hydration

4. Which activity of the debranching enzyme explains the transfer of glucose residues from a branch? [K] / [CO2]

   a). Amylo-α-1,6-glucosidase b). Glycogen phosphorylase

   c). Glycosyl 4:4 transferase d). Hexokinase

5. Select the amino acid that undergoes non - oxidative deamination by the action of dehydratase: [E] / [CO3]

   a). Serine b). Glutamic acid

   c). Alanine d). Aspartic acid

[1x5 = 5]

Part B

Fill in the Blanks questions

Answer all questions

Each question carries **2** marks

6. The synthesis of ATP from ADP and Pi is driven by ETC in a process called_____, [U] / [CO2]

7. Inhibition of SGLT-1 will mainly affect absorption of _____ and _____. [U] / [CO1]

8. Condensation reaction forms _____ ACP. [K] / [CO4]

9. The enzyme deficient in albinism is _____. [U] / [CO3]

10. The enzyme succinate thiokinase is also known as _____. [K] / [CO2]

[2x5 = 10]

Part C

Short Answer Type Questions

Answer any **5** questions

Each question carries **3** marks

11. What is glycogenesis? When is it activated? [K] / [CO2]

12. Evaluate protein digestion and assess the role of brush-border peptidases in the final stages of protein digestion. [E] / [CO1]

13. List the major pathways of lipid metabolism [K] / [CO1]

14. Explain and illustrate why urine turns dark on standing in alkaptonuria. [U] / [CO3]

15. Why is α-oxidation necessary in fatty acid metabolism? [U] / [CO4]

16. Demonstrate the action of PEP carboxylase in gluconeogenesis. [U] / [CO2]

17. Explain the role of oxidative deamination in the removal of amino groups from amino acids. [U] / [CO3]

[3x5 = 15]

Part D

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

18. Explain the reaction catalyzed by pyruvate dehydrogenase complex. [U] / [CO2]

19. Appraise the net ATP yield during the conversion of 2 molecule of pyruvate to acetyl CoA. [U] / [CO2]

20. Describe how transamination reactions contribute to the formation of glutamate. [U] / [CO3]

21. Explain why the urea cycle is considered an energy-consuming pathway. [U] / [CO3]

22. Describe the digestion of dietary triglycerides in gastrointestinal tract. [U] / [CO4]

23. Detail the stages of fatty acid oxidation. [U] / [CO4]

[5x4 = 20]